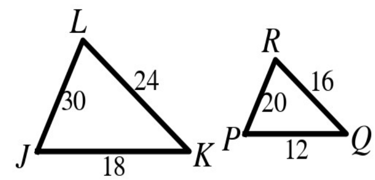
Speak **L**ike **A** **M**athematician:

similar polygons (~):

scale factor:

similarity statement:



Similar Polygons

Two polygons are similar if and only if corresponding angles are ___________________ and corresponding sides are ___________________.

Perimeters of Similar Polygons: the ratio of the perimeters equals the ratio of corresponding ___________________ (the scale factor).

Example 1 – Read a Similar Pair

In the figure at the top right, △JKL ~ △PQR.

a) Name the three pairs of congruent angles.

b) Write the three equal side ratios.

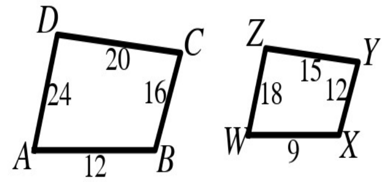
c) State the scale factor of △JKL to △PQR.

Example 2 – Are They Similar?

Quadrilaterals ABCD and WXYZ at the right have congruent corresponding angles. AB = 12, BC = 16, CD = 20, DA = 24, WX = 9, XY = 12, YZ = 15, ZW = 18.

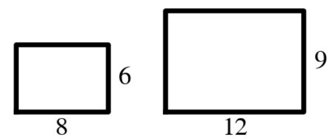
a) Check all four side ratios.

b) Write the similarity statement and scale factor.

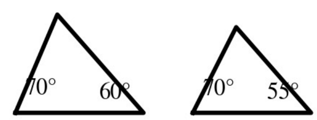


Example 3 – Similar or Not?

a) The rectangles at the right measure 8 × 6 and 12 × 9. Similar? Explain.



b) The triangles at the right have angles 70°, 60° and 70°, 55°. Similar? Explain.



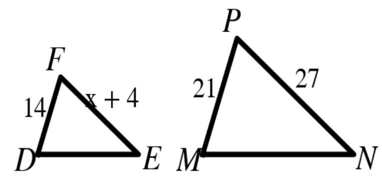
c) A classmate says ALL rectangles are similar. Correct?

Example 4 – Solve for x

In the figure at the right, △DEF ~ △MNP, DF = 14, MP = 21, EF = x + 4, and NP = 27.

a) What is the scale factor of △DEF to △MNP?

b) Write and solve the proportion for x.

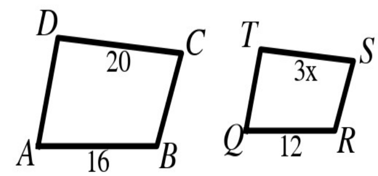


Example 5 – Quadrilateral Scale Factor

In the figure at the right, ABCD ~ QRST, AB = 16, QR = 12, CD = 20, and ST = 3x.

a) Find the scale factor of QRST to ABCD.

b) Find x.



Example 6 – Perimeters of Similar Figures

A school prints a banner similar to a 36 in. × 24 in. poster. The banner is 90 in. long.

a) Find the scale factor of the banner to the poster.

b) Find the banner's width.

c) Find both perimeters and check the ratio.